

INTRODUCTION

To gain experience in PCB design, we are to design an audio amplifier circuit that takes in and amplifies audio input, with 2W of power per audio channel as stereo output. The requirements for this project include a stereo output (sound directed through at least two speakers), turning knobs for volume/ output power control, a maximum output of 2W, LM386 model op amps as amplifier blocks, and a bridge amplifier architecture to boost the output signal. After taking note of the requirements, we had to make a list of specifications for the specific components and values needed for the circuit by doing some calculations and testing our circuit in LTSpice. When we determined through our simulations in LTSpice that our circuit met the 2W output and other requirements, we moved onto the PCB design phase, where we designed the actual physical layout of the circuit. We did not have any prior experience with Kicad, and we learned it was a bit different from LTSpice, as we had to deal with power flags, footprint assignment, and place and routes, etc. Finally after determining the correct file type to submit for the printing of our board, we were able to move on to the soldering and testing phase.

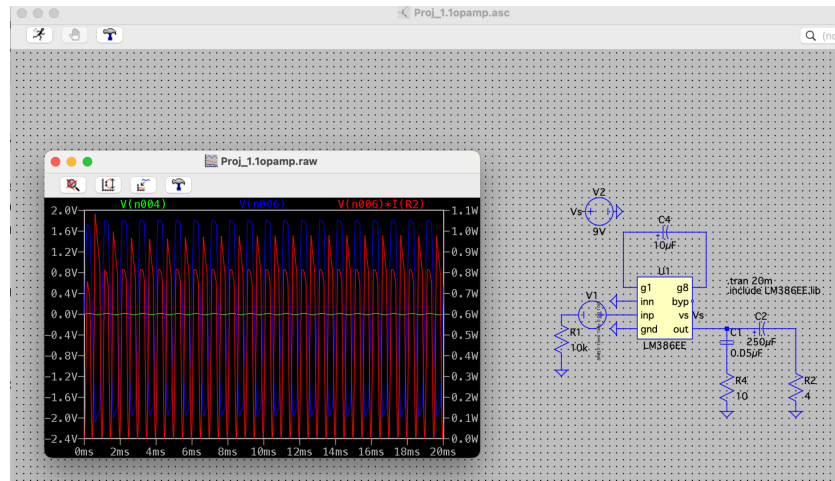
PROCESS

LTSpice Design Phase:

To ensure our circuit met the project requirements, we tested several different designs in LTSpice and observed the output waveforms. In this process, we came up with several key values to watch out for in the simulation output:

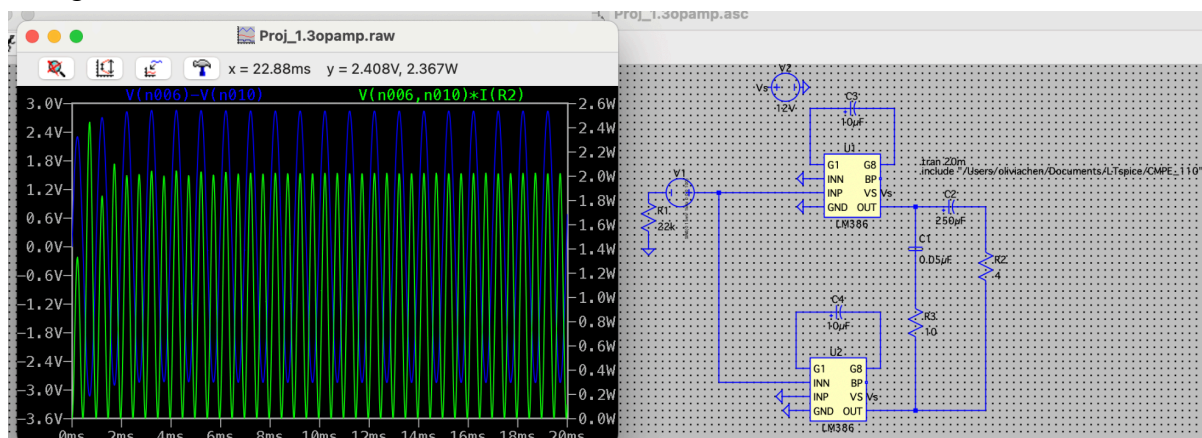
- $V_{rms} = \sqrt{PR} = \sqrt{2 * 4} = 2.828V_{rms}$
- $V_p = V_{rms} * \sqrt{2} = 4.00V$
- $V_{pp} = 8.00V$
- $I_{rms} = V_{rms}/R = 0.707 A$
 - $\rightarrow$ delivering 2W into 4ohms requires $\sim 0.7 A$ rms, or 1A peak to peak
- $V_{in,rms} = 15 mV \rightarrow$ need overall volt gain (single ended) of $2.828/0.015 = 189$ or $20*\log(189)=45.5 dB$
- LM386 GBP = 300kHz
 - Gain is almost 1 W for 1 op amp config at 1kHz

About the LM386: it is a low-voltage power amplifier, not a general op-amp, and has an internal voltage gain of 20x (26 dB) that can go up to 200x (46 dB), according to Texas Instruments' spec sheet. First, we designed and tested a one op-amp configuration, which, as expected, did not meet the requirements due to the physical limitations of the LM386.

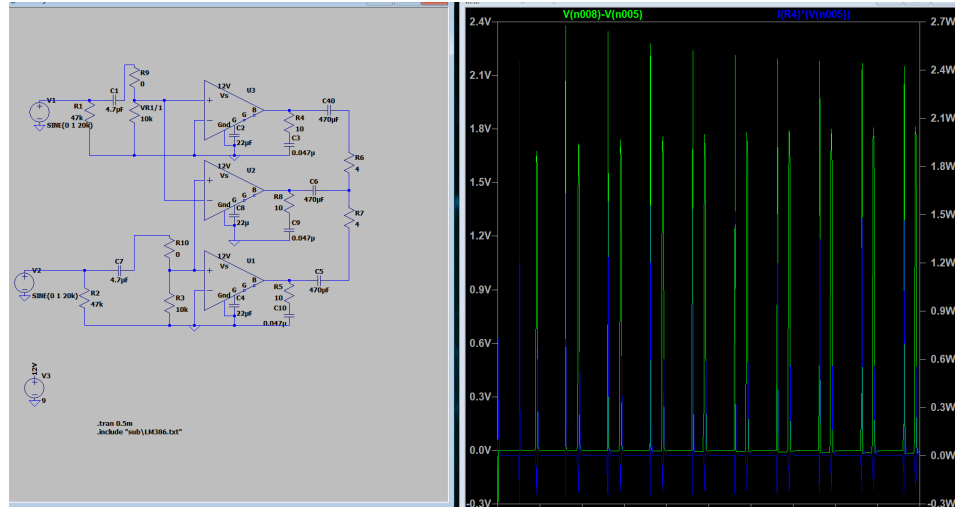


One Op-Amp Config:

As shown above, the output capability for one op amp alone is well under 2W, so a single device cannot deliver 2W into a 4ohm speaker within safe limits. Next, we tried the two op-amp configuration, with one being inverting and the other noninverting. This causes the output waves to be 180 degrees out of phase, doubling the voltage swing we saw across the previous one op-amp configuration. Shown below are the two op-amps in a bridge-tied load (BTL) configuration:



While this met the 2W output requirement, we decided to use a three op-amp configuration to most stabilize the output 2W through each 4 ohm speaker.

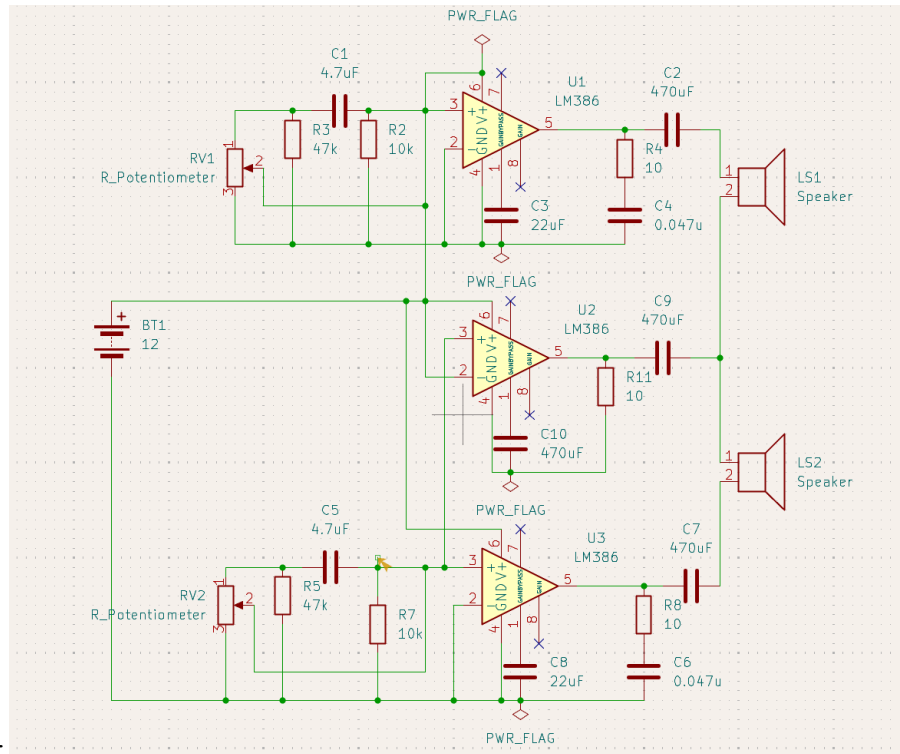


Three op-amp config:

Finally, we have the design schematic of our circuit that has an appropriate simulated output. Next, we replicated our LTSpice schematic design in KiCad to begin designing the physical layout and materials of our actual PCB board.

PCB Design Phase:

We completed the KiCad schematic and the calculations needed for the component values, making sure it has passed the ERC checker; thus, we proceeded to footprint placements. The footprint is the PCB representation of a component, including its pads, outline, silkscreen, and 3D model, and is needed by each component in the finished design of the PCB board. Pads are the exposed regions of metal on a circuit board that the component lead is soldered to and there are two types available: through-hole and surface mount pads. Pads with through-holes will have their holes coated with protective plating, which is done using the process of electrolysis. Meanwhile, surface mount pads are used for single-sided boards and have areas around the unplated holes that are clear of any copper in order to prevent shorts between the copper layers and the components. We did not have to create any custom footprints, as the necessary ones were given to us below for the 3 op-amp circuit design.



KiCad Schematic:

Footprint Assignment: As shown above, we are using through hole (THT) footprints for our component parts, meaning we inserted our PCB components through the drilled holes in the board. Generally for THT, components are larger (footprints are also bigger), making manual assembly and rework easier at the cost of more board space. We matched each symbol in the schematic to the listed below footprints using KiCad's *Assign Footprints* tool.

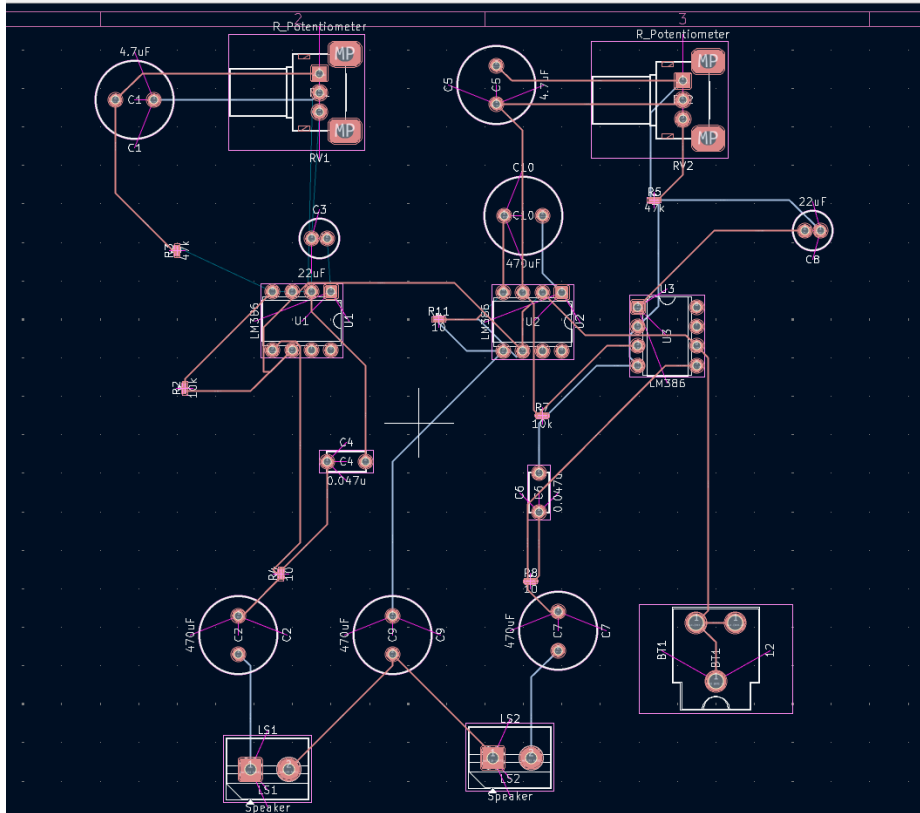
Symbol : Footprint Assignments

```
1      C1 -          470 uF : Capacitor_THT:C_Radial_D10.0mm_H16.0mm_P5.00mm
2      C2 -          22 uF : Capacitor_THT:C_Radial_D5.0mm_H11.0mm_P2.00mm
3      C3 -          0.047 uF : Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
4      C4 -          470 uF : Capacitor_THT:C_Radial_D10.0mm_H16.0mm_P5.00mm
5      C5 -          22 uF : Capacitor_THT:C_Radial_D5.0mm_H11.0mm_P2.00mm
6      C6 -          0.047 uF : Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
7      C7 -          470 uF : Capacitor_THT:C_Radial_D10.0mm_H16.0mm_P5.00mm
8      C8 -          470 uF : Capacitor_THT:C_Radial_D10.0mm_H16.0mm_P5.00mm
9      C9 -          0.047 uF : Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
10     C10 -         22 uF : Capacitor_THT:C_Radial_D5.0mm_H11.0mm_P2.00mm
11     C11 -         4.7uF : Capacitor_THT:C_Radial_D10.0mm_H16.0mm_P5.00mm
12     C12 -         4.7 uF : Capacitor_THT:C_Radial_D10.0mm_H16.0mm_P5.00mm
13     D1 -          LED : LED_THT:LED_D5.0mm
14     J4 - Screw_Terminal_01x02 : TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
15     J5 - Screw_Terminal_01x02 : TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
16     Power1 - Screw_Terminal_01x02 : TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
17     R1 -          1k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
18     R2 -          10 : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
19     R3 -          10 : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
20     R4 -          10 : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
21     R5 -          47k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
22     R6 -          47k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
23     RV1 -         10k : Potentiometer_Knob_vertical_THT:pot_vertical_Flutesan_9_7mm_P2_5mm
24     RV2 -         10k : Potentiometer_Knob_vertical_THT:pot_vertical_Flutesan_9_7mm_P2_5mm
25     Speaker_L1 - Screw_Terminal_01x02 : TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
26     Speaker_R1 - Screw_Terminal_01x02 : TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
27     SW1 -          SW_SPST : Button_Switch_THT:SW_Slide-03_Wuerth-WS-SLTV_10x2.5x6.4_P2.54mm
28     U1 -          LM386 : Package_DIP:DIP-8_W7.62mm
29     U2 -          LM386 : Package_DIP:DIP-8_W7.62mm
30     U3 -          LM386 : Package_DIP:DIP-8_W7.62mm
```

Component Placement: After matching all the schematic components to foot prints, we were able to transfer the board to the PCB layout editor, where we placed all the components in the order of: inputs → gain stage → output stage → load. We centered the LM386 op-amps to minimize trace lengths, placed high-current output capacitors near the speaker screw terminals, and placed potentiometers near the board edge for convenient adjustment.

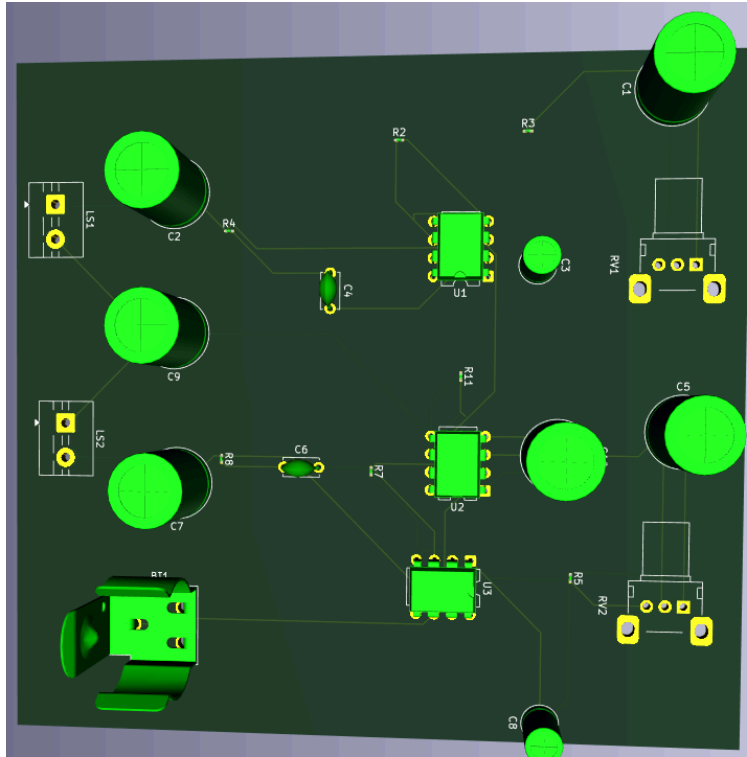
Routing:

We routed +Vcc using widened traces suitable for high current, organized ground nets to minimize loop area and noise, and moved some routing to the back layer to reduce crossovers. In addition, we tried keeping sensitive input traces short, the gain-setting components tightly routed around the op-amp pins, and the output path traces wide so as to support speaker current. Routing was completed until all nets were connected without design rule violations and passed KiCad's *Design Rule Checker* (DRC).



3D Board Rendering:

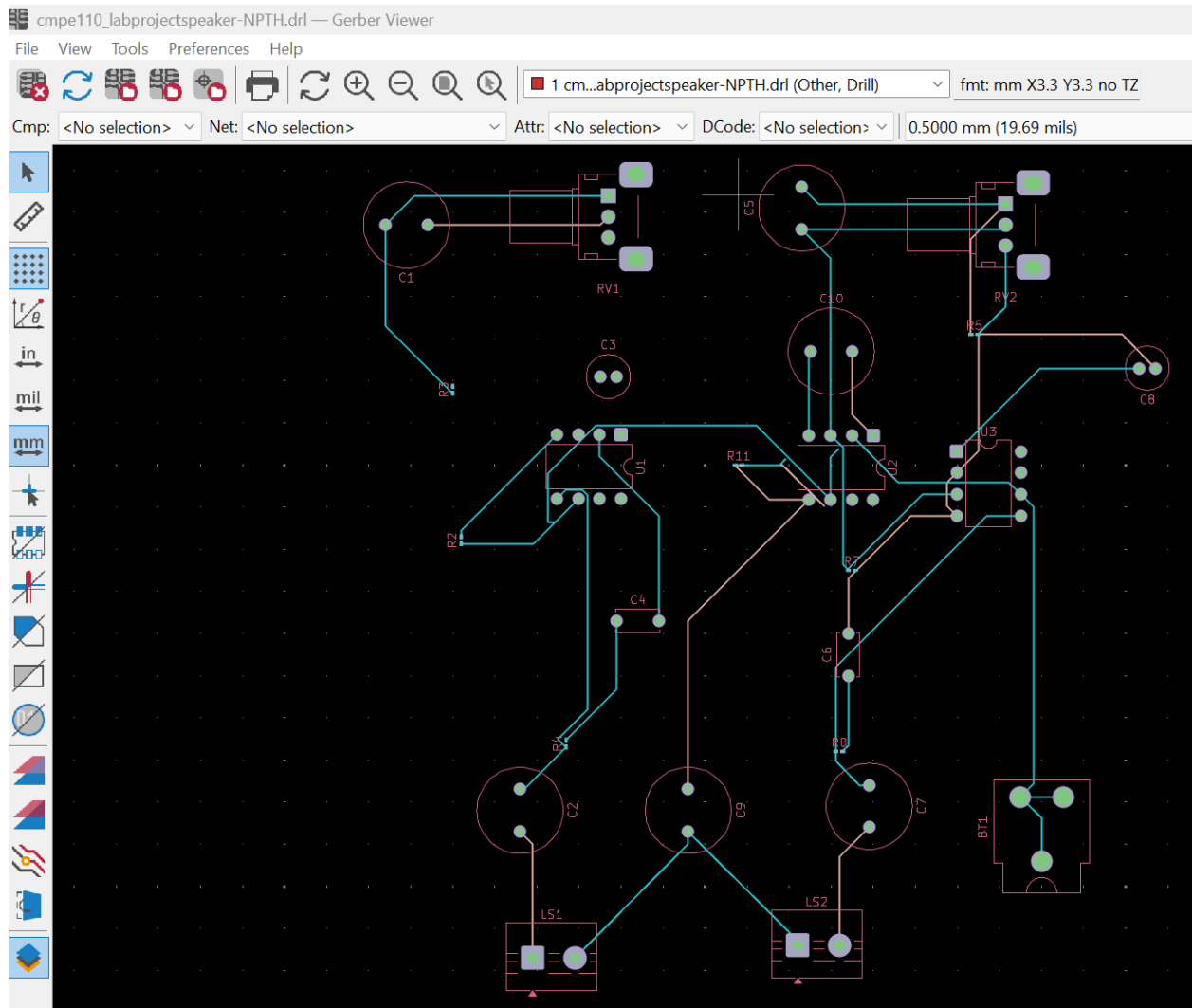
For the final step of the KiCad designing process is the 3D board rendering, where we generated a full 3D view of the PCB, confirming that all components have valid footprints, orientation and spacing are correct, and through-hole components align properly with the board geometry. In our next steps, we gathered all the required components for the physical board (LM386 op-amps, capacitors, resistors, etc), soldered the components onto the PCB with the proper orientation, and inspected all solder joints for bridges and cold joints. We learned that manufacturing companies don't accept KiCad project files and instead require "universal" manufacturing files called Gerbers, which are like the blueprints (copper, masks, silkscreens, board outline) for each PCV layer, as well as drill files, which define hole diameters and positions. Once we were done with mounting the components cleanly and soldering, we did a continuity check with a multimeter to verify that each component pad has been properly connected.



Bill of Materials (BOM)

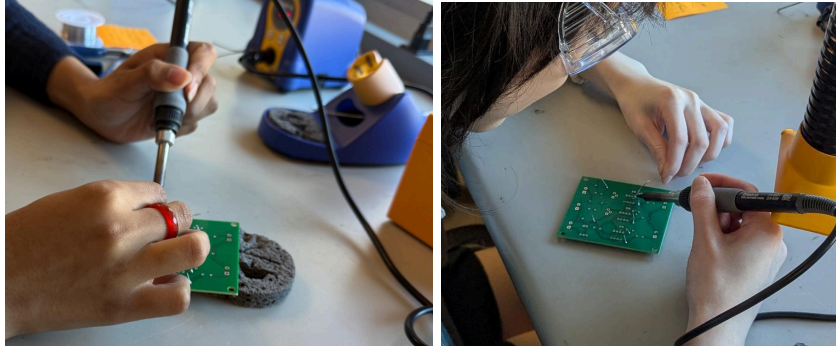
	A	B	C	D	E	F	G
1	Designator	Value	Footprint	Supplier	Part_Number	Unit_Price_USD	Notes
2	C1	470 μ F	C_Radial_D10.0mm_H16.0mm_P5.00mm	Generic	Electrolytic_470uF_10V	0.3	Radial electrolytic capacitor
3	C2	22 μ F	C_Radial_D5.0mm_H11.0mm_P2.00mm	Generic	Electrolytic_22uF_100V	0.14	Radial electrolytic capacitor
4	C3	0.047 μ F	C_Disc_D5.0mm_W2.5mm_P5.00mm	Generic	Ceramic_Disc_47nF_50V	0.11	Ceramic disc capacitor
5	C4	470 μ F	C_Radial_D10.0mm_H16.0mm_P5.00mm	Generic	Electrolytic_470uF_10V	0.3	Radial electrolytic capacitor
6	C5	22 μ F	C_Radial_D5.0mm_H11.0mm_P2.00mm	Generic	Electrolytic_22uF_100V	0.14	Radial electrolytic capacitor
7	C6	0.047 μ F	C_Disc_D5.0mm_W2.5mm_P5.00mm	Generic	Ceramic_Disc_47nF_50V	0.11	Ceramic disc capacitor
8	C7	470 μ F	C_Radial_D10.0mm_H16.0mm_P5.00mm	Generic	Electrolytic_470uF_10V	0.3	Radial electrolytic capacitor
9	C8	470 μ F	C_Radial_D10.0mm_H16.0mm_P5.00mm	Generic	Electrolytic_470uF_10V	0.3	Radial electrolytic capacitor
10	C9	0.047 μ F	C_Disc_D5.0mm_W2.5mm_P5.00mm	Generic	Ceramic_Disc_47nF_50V	0.11	Ceramic disc capacitor
11	C10	22 μ F	C_Radial_D5.0mm_H11.0mm_P2.00mm	Generic	Electrolytic_22uF_100V	0.14	Radial electrolytic capacitor
12	C12	4.7 μ F	C_Radial_D10.0mm_H16.0mm_P5.00mm	Generic	Electrolytic_4.7uF_35V	0.05	Radial electrolytic capacitor
13	R1	1 k Ω	Resistor_THT_Axial:DIN0207_6.3mm_D2.5mm_P10.16mm	Generic	Resistor_1k Ω _MF	0.12	Metal-film resistor
14	R2	10 Ω	Resistor_THT_Axial:DIN0207_6.3mm_D2.5mm_P10.16mm	Generic	Resistor_10 Ω _MF	0.1	Metal-film resistor
15	R3	10 Ω	Resistor_THT_Axial:DIN0207_6.3mm_D2.5mm_P10.16mm	Generic	Resistor_10 Ω _MF	0.1	Metal-film resistor
16	R5	47 k Ω	Resistor_THT_Axial:DIN0207_6.3mm_D2.5mm_P10.16mm	Generic	Resistor_47k Ω _MF	0.1	Metal-film resistor
17	R6	47 k Ω	Resistor_THT_Axial:DIN0207_6.3mm_D2.5mm_P10.16mm	Generic	Resistor_47k Ω _MF	0.1	Metal-film resistor
18	RV1	10 k Ω	Potentiometer_Knob_vertical_THT: pot_vertical_Flutesan_9_7mm_P2.5mm	Alps-Alpine	RK09K1130AH1	1.31	Vertical 10 k Ω pot
19	RV2	10 k Ω	Potentiometer_Knob_vertical_THT: pot_vertical_Flutesan_9_7mm_P2.5mm	Alps-Alpine	RK09K1130AH1	1.31	Vertical 10 k Ω pot
20	U1	LM386	Package_DIP-8_W7.62mm	Texas Instruments	LM386N-4/NOPB	1.05	Audio amplifier IC
21							
22							
...							
						Total 6.19	

Gerber File:



Soldering and Testing:

After gathering all the components needed, we began to solder them to the printed board. We set the soldering iron to about 720F and applied a bit of solder to the tip to help the solder stick, and we made sure to clean the tip of the iron after each connection. During the soldering process, we found that if we lingered too long with the iron, the plastic on the PCB board would be damaged, so we needed to be quick and lighthanded. Soldering the op-amps was the most difficult part for us, as the legs are quite close together and we had to be careful not to create any shorts.

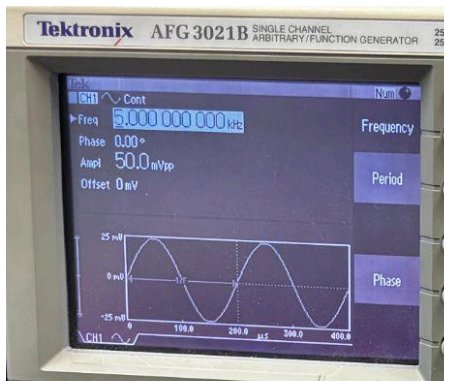
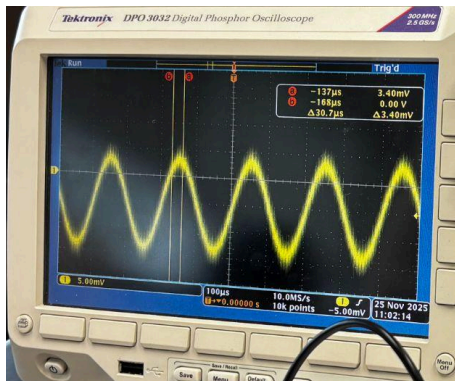
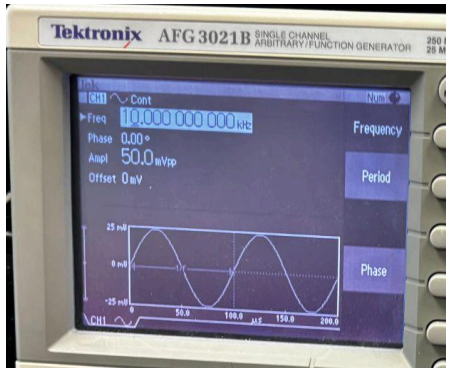
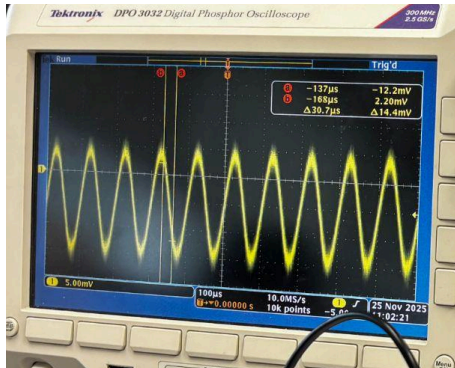
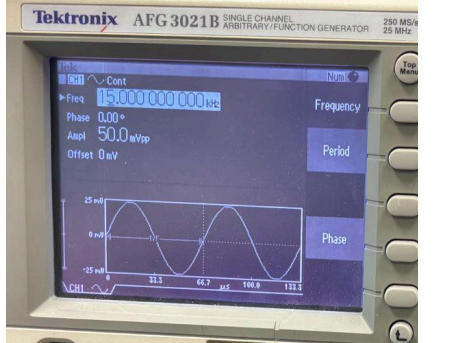
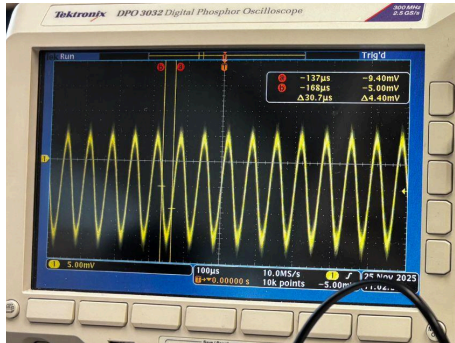
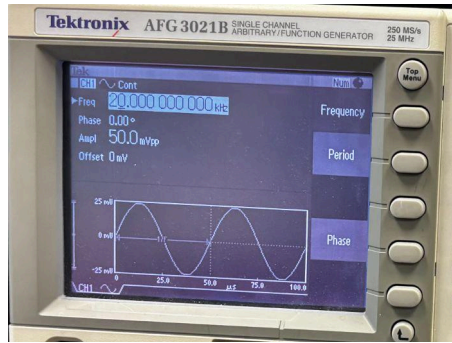
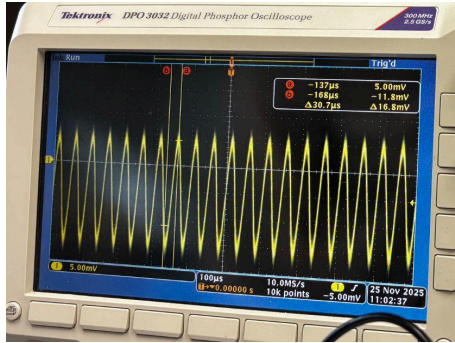


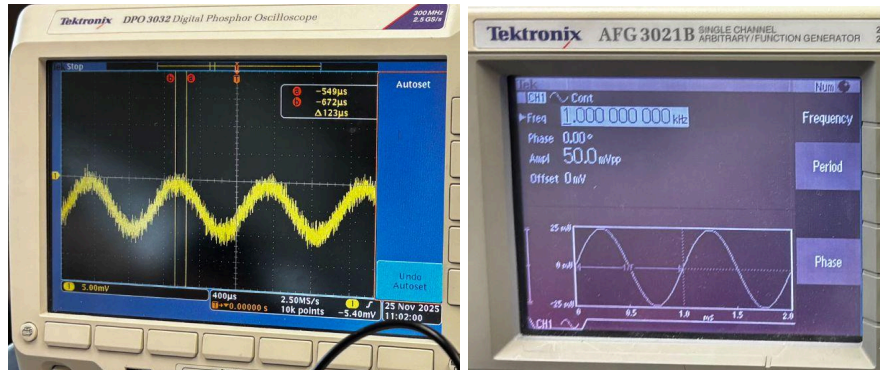
Finished PCB Board:



PCB Testing:

For our testing phase, we connected power input to a 9V DC power supply, the audio input to a 1kHz-0.3V-1ohm function generator, and the output to a speaker. We tested the output sound across varying frequencies, and though our initial reading on the oscilloscope connected to the speaker was quite noisy, we were able to fix it by adjusting our oscilloscope probe settings and resoldering some of the op-amps and capacitors.





REFERENCES:

https://www.digikey.com/htmldatasheets/production/3514971/0/0/1/lm386.html?psafe_param=1&gclsrc=aw.ds&gad_source=1&gad_campaignid=9265913509&gbraid=0AAAAADrbLljOhAxpKTr7FvLUb2shzvn5s&gclid=CjwKCAjwr8LHBhBKEiwAy47uUieydn0vqFeNyVofWluYyg69SYgJ0CSBl_aEKrjkvc6VKM5NMgd7mhoCz0AQAvD_BwE